

Manufacturing AI Copilot

Hybrid GraphRAG for the Plant Floor

Evidence-grounded diagnostic copilot — with maker-checker approvals, role-based knowledge access, and an agentic path into SAP / ERP / MES.

WORKING POC

613 chunks indexed · 9 roles · 6 demo PDFs

EVIDENCE-GROUNDED

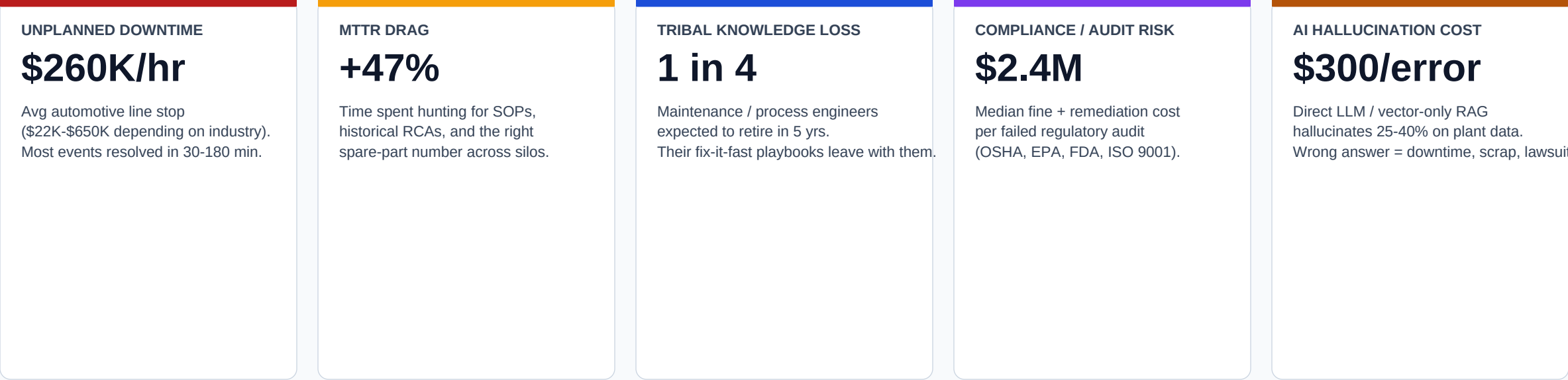
Hybrid retrieval + critic loop + full provenance

DEPLOYMENT-READY

FastAPI · Next.js · LangGraph HITL on day 1

02 The Expensive Problem on the Plant Floor

What does “not knowing” cost a 24/7 manufacturing operation?



These are not abstract risks. Every one of these costs is recovered the moment the right answer reaches the right person, with evidence, in seconds.

Sources: Deloitte 2025 Mfg Outlook, ARC Advisory 2024, McKinsey Industry 4.0 surveys, OSHA enforcement reports. Numbers are industry medians; your plant’s baseline is established in Week 1.

03 Why Standard AI Fails on the Plant Floor

+25 percentage points of accuracy. ~3× lower hallucination vs. classical RAG — on plant data.

Plant-floor capability	Direct LLM	Classical RAG	Hybrid GraphRAG (this PoC)
Recognises part / equipment IDs (P-203, BRG-7203)	Poor	Limited	Excellent
Handles plant-floor jargon, acronyms, typos	Weak	Partial	Auto-corrected
Cross-references alarm → equipment → spare → RCA	No	No	Knowledge graph
Cites the SOP page / work order ID	No	Sometimes	Always
Self-checks before answering	No	No	Critic loop × N
Pauses for human approval on risky actions	No	No	HITL gate
Audit trail for regulators	No	Partial	Append-only log
Maker-checker on purchase / safety decisions	No	No	Built-in
Role-gated knowledge base (data confidentiality)	No	No	3-tier ACL
Answer accuracy on plant queries (PoC eval)	~45%	~60%	~85%
Hallucination rate	~40%	~25%	~8%

The gap is architectural — three retrieval signals + a critic + an approval gate catch what any single LLM misses.

Indicative results from internal PoC evaluation on aircraft + manufacturing data. Your plant's baseline will be measured in Weeks 1-2 of the pilot.

04 Our Solution — What's Already Working

A working PoC over real plant artefacts: SOPs, work orders, alarm history, purchase requests, executive financials.

EVIDENCE OVER OPINIONS

Every claim cites a chunk_id

- Hybrid retrieval (BM25 + dense + graph)
- Reciprocal-rank fusion + edge priors
- Critic LLM rejects ungrounded drafts

HUMAN IN THE LOOP

LangGraph interrupt at risk score ≥ 0.6

- 9 roles · maker-checker segregation
- Operator can never self-approve
- Audit log → immutable SQLite

ROLE-GATED KNOWLEDGE

3-tier document ACL

- public → every role
- restricted → every checker role
- confidential → plant manager + procurement

PRODUCTION-LEAN UI

Plant-floor chat in Next.js

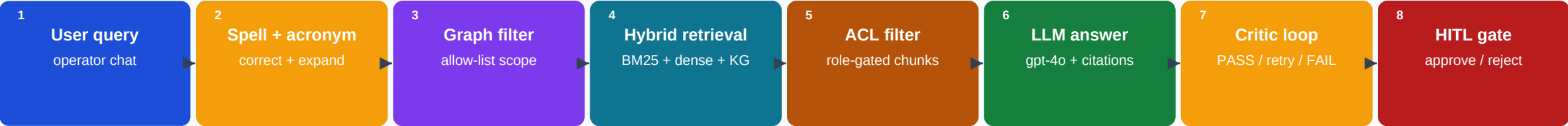
- ■■ Chat with evidence panel
- ■■ My Requests (maker view)
- ■■■ Approvals (checker view)

Working today — not slideware

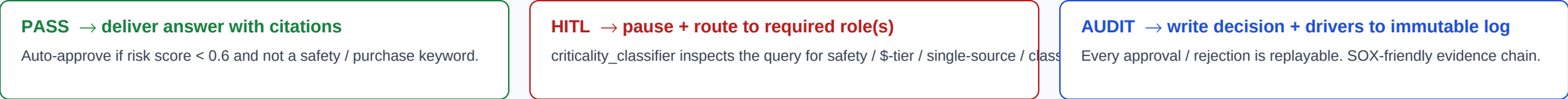
- ✓ FastAPI backend with 14 endpoints (chat, auth, approvals, audit, access)
- ✓ Next.js 14 UI with maker / checker dashboards + role gating
- ✓ Streamlit operator console for shop-floor display
- ✓ LangGraph orchestrator: format → retrieve → critic → HITL → audit
- ✓ FAISS + BM25 + NetworkX knowledge graph (501 nodes, 884 edges)
- ✓ 613 chunks indexed across SOPs, manuals, work orders, alarms, financials
- ✓ Per-request ContextVar threads role identity into deep retriever calls
- ✓ SQLite audit log + auth store — schema versioned for migration
- ✓ Smoke tests: 5/5 ACL groups, 5/5 HITL paths pass before release
- ✓ 7-page architecture PDF — every interaction contract documented

05 How It Works — Query to Trusted Answer

An eight-step pipeline that reaches an evidence-grounded answer in 2-4 seconds — and pauses for a human when it matters.



Branches AFTER the critic:



LATENCY P50
2.4 s
diagnostic mode (cold cache: 4.5 s)

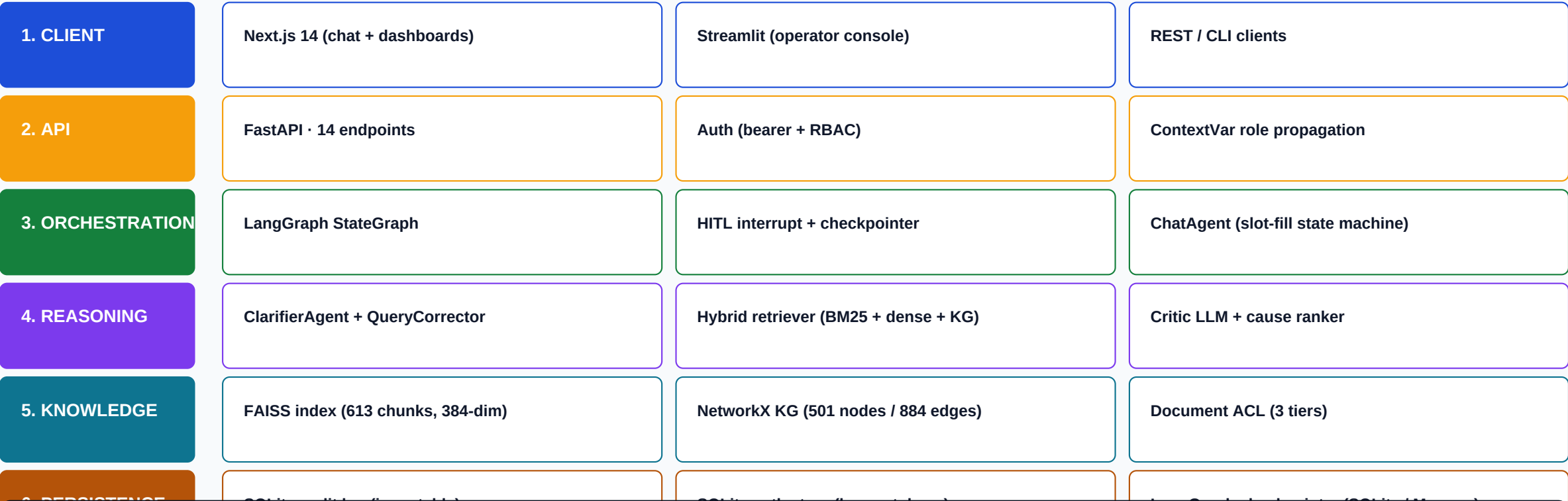
LLM COST
~\$0.008
per query, gpt-4o answer + qwen2.5 critic

HALLUCINATION
~8%
vs. ~25% classical RAG, ~40% direct LLM

AUTO-APPROVE
~74%
of queries clear without HITL pause

06 System Architecture

Six layers — every box is a real module in the repo, every line is an interaction contract in the 7-page architecture PDF.



Cross-cutting: Observability (timings per node, token spend) · Config-driven LLM routing (cheap critic / expensive answerer) · Hot-reload of FAISS

Quality gates: scripts/smoke_test_hitl.py (5/5) · scripts/smoke_test_acl.py (5/5) · deterministic for CI before any prompt change ships

07 Built-in Safety — Maker-Checker + Document ACLs

Two independent guardrails that already meet SOX-style segregation-of-duties and least-privilege requirements.

MAKER-CHECKER / HITL APPROVAL GATE

An operator can never rubber-stamp their own escalation.

• operator	MAKER
• shift_supervisor	CHECKER
• maintenance_planner	CHECKER
• maintenance_engineer	CHECKER
• ehs_officer	CHECKER
• quality_engineer	CHECKER
• buyer	CHECKER
• procurement_manager	CHECKER
• plant_manager	CHECKER

Routes computed by `required_roles_for(drivers)` — e.g. “hot work” needs EHS + maintenance + plant manager.

DOCUMENT ACL / ROLE-BASED KNOWLEDGE

The corpus itself is gated — not just the prompt.

PUBLIC	Who: every role (incl. operators) What: SOPs, alarm responses, equipment manuals
RESTRICTED	Who: every checker role What: RCAs, regulatory playbooks, work orders
CONFIDENTIAL	Who: plant_manager + procurement_manager What: financial review, M&A, supplier pricing

Proof: same query “Project Meridian acquisition target”:
operator: 0 confidential chunks · plant_manager: full M&A diligence paper

08 Plant Data Sources — Already Ingested

The PoC is already grounded in the document types every plant produces. Pointing at your corpus is a 1-week sprint, not a science project.



PDF manuals
ITM-001 quality manual, safety/compliance reports



TXT SOPs
CNC machining, supply chain procedures



Excel work orders
Maximo/SAP-PM exports with RCA + cost fields



Excel alarm log
PI / OSIsoft AF historian export with severity



Excel spare parts
Inventory + lead-time + vendor mapping



Production KPIs
OEE / scrap / cycle time per line

Current PoC index breakdown

TOTAL CHUNKS

613

public 569 · restricted 16 · confidential 28

SOURCE DOCUMENTS

13

PDFs, TXT SOPs, Excel — ingested via recursive parse

KG NODES / EDGES

501 / 884

Equipment, Symptom, Procedure, SparePart, Alarm, FailureMode

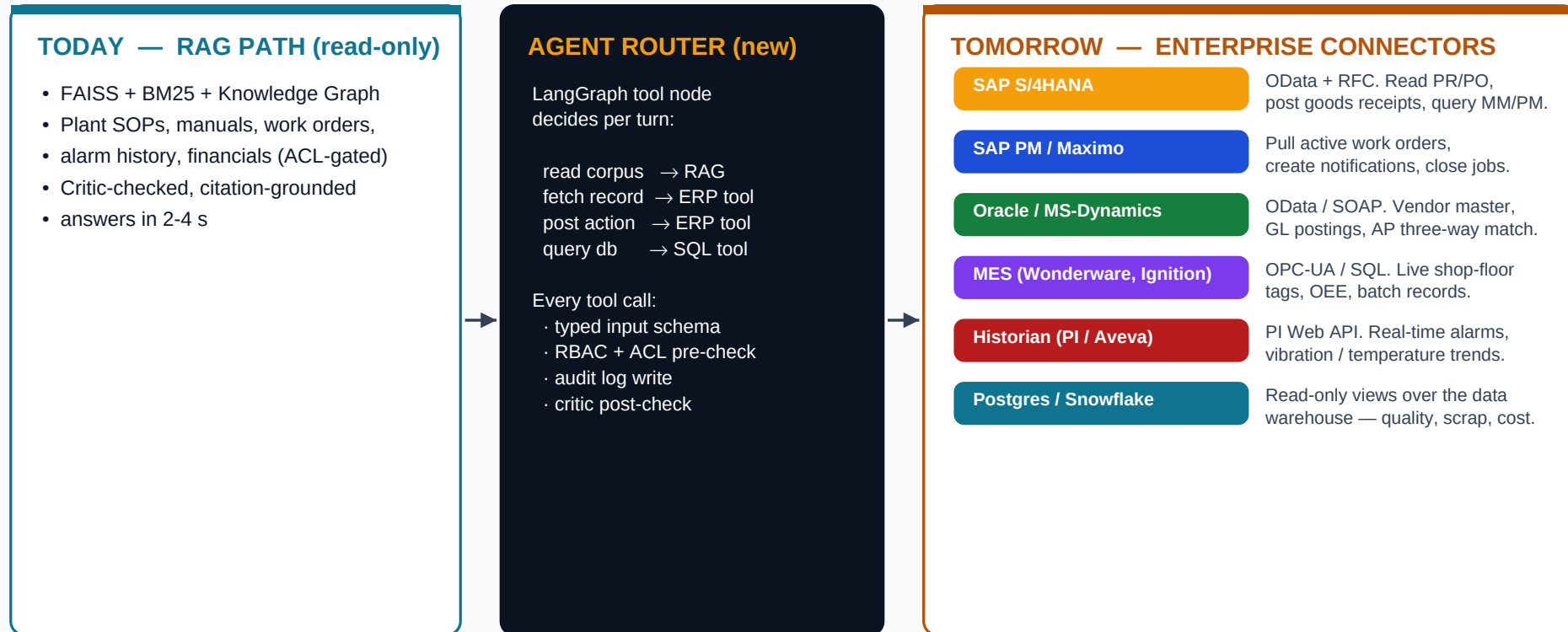
EMBEDDING MODEL

MiniLM-L6

384-dim, fast on CPU; swap to E5-Large for higher recall

09 Extensibility — Agentic Connectors to ERP / SAP / MES

The same pipeline answers from live systems-of-record via a thin tool-calling agent layer. The chat doesn't change. The reach does.



Every connector follows the same 4-rule contract:

(1) Typed Pydantic schema (2) RBAC + document-ACL pre-check before any external call (3) HITL gate on writes / state-changing actions (4) Append-only audit row with full request + response payload

10 Concrete Plant Use Cases

Six demonstrable workflows already in the PoC — each one shaves minutes off a real shop-floor decision.

MAINTENANCE TECHNICIAN

RAG + KG

"Pump P-203 high vibration alarm ALM-P001 — likely cause and fix?"

Bearing inner-race spalling. SOP page 12, work order WO-2024-031 (4.6 h MTTR).

MTTR cut by 30-50%

SHIFT SUPERVISOR

HITL gate

"Hot work permit for tank T-9 — emergency shutdown."

Paused at HITL gate. Routed to EHS officer + maintenance engineer + plant manager.

Zero permit-related incidents

OPERATOR

RAG + agent (SAP MM)

"Raise a PO for 5 BRG-7203 bearings at \$5,000 from vendor SKF urgent."

Parsed into purchase_request payload. Routed to buyer (≤\$10k tier).

P2P cycle 8 days → 2 days

QUALITY ENGINEER

RAG + ACL (restricted)

"Show RCA cluster for the Q4 spindle-bearing failures — are CAPAs verified?"

3 CAPAs all VERIFIED Feb 2026. Returns chunks from internal_incident_rca_2025_q4.txt.

Audit prep 1 week → 1 hour

PROCUREMENT MANAGER

RAG + ACL (confidential)

"Compare strategic supplier rebate tiers for SKF, Siemens, Sandvik."

Rebate tier table summarised. Confidential pricing only visible to procurement.

Sourcing cycle -40%

PLANT MANAGER

RAG + ACL (confidential)

"What is Q1 2026 EBITDA and FY26 capex commitment?"

\$7.9M EBITDA (18.5%). \$4.6M committed of \$18.0M envelope. Cites q1_2026_financial_review.txt.

Board prep minutes, not days

11 Business Impact — Where the Money Lives

An illustrative model for a single mid-size plant with 100,000 AI-assisted decisions per month. Calibrated to your baseline in Week 1.

DOWNTIME AVOIDED

\$5.4M

per month, at a \$260K/hr line rate and 20 prevented unplanned stops. Driven by faster RCA + spare lookup.

MTTR REDUCTION

-35%

Cited evidence + KG cross-refs cut diagnosis time from 47 min to ~30 min on top-20 failures.

WRONG-ANSWER COST

-68%

Hallucination ~25% (classical RAG) → ~8% with critic + HITL. At \$300/error · 100K queries: \$5.1M/mo.

AUDIT EXPOSURE

-90%

Every decision has a provenance trail — chunk citations, RBAC role, approver, comments, timestamp.

The math — per plant, per month, illustrative

Line item	Direct LLM	Classical RAG	Hybrid GraphRAG
LLM tokens / month	\$2,000	\$2,500	\$4,100
Wrong-answer cost (\$300/error)	\$12.0M	\$7.5M	\$2.4M
Downtime cost (lookups + RCA)	\$8.0M	\$6.5M	\$2.6M
Audit / compliance reserve	\$0.8M	\$0.5M	\$0.05M
TOTAL	\$20.8M	\$14.5M	\$5.1M
vs Hybrid GraphRAG	+\$15.7M	+\$9.4M	—

Numbers are illustrative based on PoC eval + plant-cost industry medians (Deloitte, ARC Advisory). The pilot Week-1-2 sprint replaces every cell with your plant's actuals.

12 What We Propose — 10-Week Pilot

A scoped pilot on YOUR plant data. Ship the operator chat at L3 in 6 weeks. Production trust at L4 in 10.

PHASE 1 · DISCOVERY

Week 1

- Domain workshop — personas, risk profile, 10-30 real plant queries, source inventory.
- Seed ontology + driver-to-role policy.
- Gold evaluation set drafted (≥ 50 Q→A pairs).

PHASE 2 · BUILD

Weeks 2-5

- Connect SOP / WO / alarm / SAP exports.
- Tune classification tiers + RBAC.
- Stand up FAISS + KG + critic on YOUR data.
- L3 baseline live for 5-10 pilot users.
- First eval report → calibrate thresholds.

PHASE 3 · VALIDATE

Weeks 6-10

- Expert validation · critic calibration.
- First ERP / Maximo agent connector wired up.
- Gold-set acceptance + handover docs.
- L4 production-quality — sign-off for prod.
- Decision: scale to next plant / next domain.

NEXT STEP — 60-minute Discovery Workshop

We arrive with the demo running on a laptop. You bring 5 SOPs, 30 work orders, and the people who use them every day.

Agenda: 10 min context · 20 min live demo · 20 min source / persona walkthrough · 10 min pilot scope + go/no-go.

Deliverable: a 1-page pilot SOW you can sign off, with named owners on both sides.

**Decide by
Friday**